

Broadcast :

devices in the
sending data to all the network

e.g.

teacher → entire class

Multicast :

Sending data to a specific group
of devices.

e.g. teacher → science club students

Collision domain :

It is an area where a collision
can occur.

where collision happen ?

* Hub

* 10Base2, 10Base5 (old Bus)

why?

* They share one common communication channel.

where do not ?

* Switch (each port is separate)

* Router

Broadcast domain :

It is an area where broadcast message reaches.

eg: if one device sends a broadcast, all devices in the same broadcast domain receive it.

Devices that extend broadcast domain:

- * Switches

- * Hubs

Devices that break broadcast domain -

- * Routers

- * Layer 3 devices.

CSMA (Carrier sense multiple access) :

CSMA is a method where devices listen before sending.

Simple meaning:

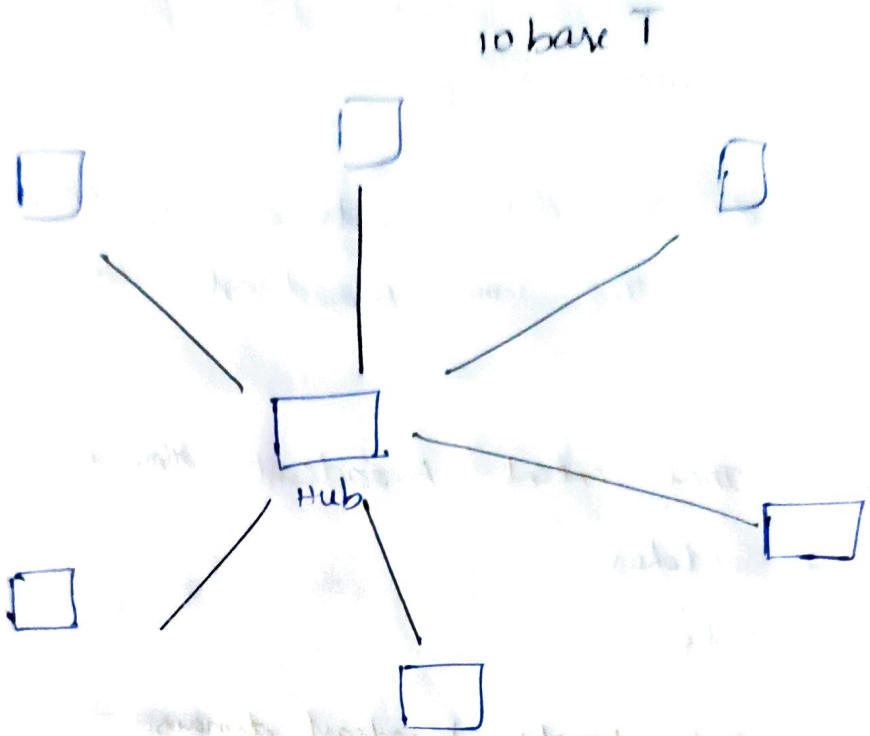
check if the line is free, then send.

CD (Collision detection) :

used in half-duplex Ethernet (Hubs).

- * if a collision occurs, devices stop, wait for a random time and resend.

Star Topology



* Star topo. with hub $\Rightarrow$ Physically Star
Logically bus

Reasons:

The hub repeats the signal from a device to every port.

• hub

* Works only at physical layer

* There one collision domain

* shared bandwidth

* Broadcast to everyone

Star with switch:

* Physically & Logically star.

* switch knows the device's MAC address & know where they are whereas hub doesn't know this.

Key points:

* Bandwidth = Maximum capacity of a communication link.

* Throughput = Actual achieved rate

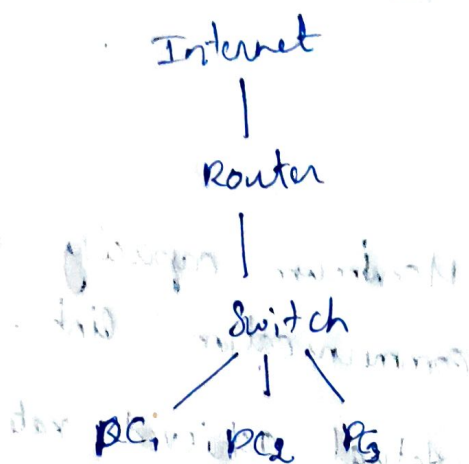
Summary:

Hub	Bridge	Switch
Layer 1	Layer 2	Layer 2
No MAC learning	Learns MAC addr.	Learns MAC addr.
forwards to all ports	forwards only to destination port	same as bridge
1 collision domain	1 collision domain per port	Same as bridge
1 broadcast domain	1 broadcast domain	1 broadcast domain (unless VLANs are used)
Software	Software	Hardware

Routed Network:

1) Why we need a router:

Suppose you have a home network



* PCs communicate inside your home

* The router connects your home to internet

Switch → Connects devices within the same network (LAN)

Router → Connects different networks

2) LAN ?

A is network inside a limited area such as

- Home
- office
- college
- school

3) WAN ?

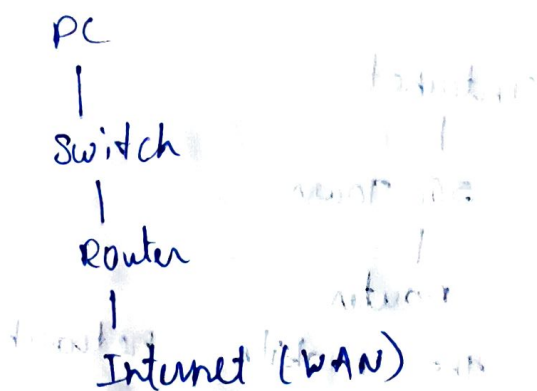
It connects different LANs over large distances

Home LAN ---- Internet ---- Office LAN

* The internet is the world's largest WAN

4) Router connects LAN to WAN

Home LAN :



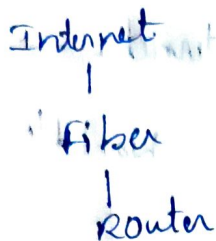
* The router has one interface toward your LAN.

* Another interface toward your ISP / Internet.

So it connects 2 different Networks

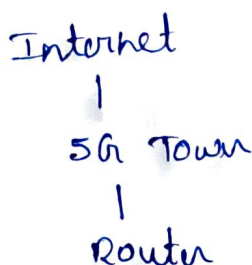
5) Different ways a router connects to the internet.

i) Fiber



* Fastest & most common today

ii) 5G Router



* uses the mobile network

iii) 4G Router

same idea but using 4G LTE

iv) Starlink



Satellites
uses instead of cables

v) DSL / ADSL

Older Technology

Telephone Line

|

Router

* uses telephone wire

6) why Add a switch?

Suppose your router has only 4 LAN ports.

But your office has 40 computers.

* The switch gives you many more ports.

7) Company with Two offices

Office A

PCs

|

Switch

|

Router

Office B

PCs

|

Switch

|

Router

connected through internet

LAN A — Router A — Internet — Router B — LAN B

* Now an employee in Office A can communicate with Office B.

* This is called routing.

8) What does 'Routing' mean?

choosing a path to send packets from one network to another.

9) Routers Don't forward broadcasts.

* If a PC sends an ethernet broadcast the switch forwards it to everyone in the LAN.

* But when it reaches the router.

* The router stops it. It does not send it to the internet.

Reason:-

Millions of unnecessary broadcast frames would flood the internet Network.

10) router = single separate broadcast domain

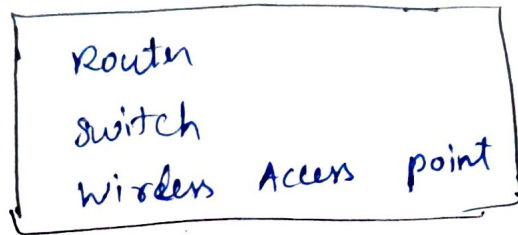
* Each router Interface creates a new broadcast domain.

11) collision domain

each router interface is also a separate collision domain.

12) Modern home routers are more than just routers.

Home router



* In a typical home router, the LAN ports are one single broadcast domain because they are internally connected by a switch.

* They are not four separate LANs, even though they're labelled as LAN1, LAN2, LAN3, & LAN4